Marcel Neunhoeffer

Contact
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Career Path

Academic Employments

Institute for Employment Research, Nuremberg
Postdoctoral Research Associate, Statistical Methods Unit
Since 2023

Ludwig-Maximilians-Universität München
Teaching and Research Associate, Statistics and Data Science for the Social Sciences and Humanities
Since 2021

Diplomatische Akademie Wien
Lecturer - Machine Learning
2025–2026

Boston University
Research Associate, Computer Science
2021–2023

University of Mannheim
Teaching and Research Associate, Quantitative Methods in the Social Sciences
2016–2021

University of California, Berkeley
Lecturer - Information and Cybersecurity, School of Information
2020

Non-Academic Employments

Staatsministerium Baden-Württemberg
Project Staff, Policy and Strategy Department
2016

Bündnis 90/Die Grünen
Campaign Staff in the Federal Office Berlin, and the Regional Office Berlin
2013, 2014

Education

University of Mannheim
Ph.D. Political Science (Dr. rer. soc.)
Graduate School of Economic and Social Sciences
Thesis: Generative Adversarial Nets for Social Scientists
Defense Date: March 31, 2023

University of California, Berkeley
Visiting Ph.D. Student, Simons Institute for the Theory of Computing
2019

University of Mannheim
M.A. Political Science
2016

University of Passau
B.A. Governance and Public Policy
2013

Research

Peer-Reviewed Articles

16. **Marcel Neunhoeffer**. Forthcoming. An Introduction to Generative Adversarial Nets in R: The RGAN package, in: *R Journal*.

15. **Marcel Neunhoeffer**, Jeremy Seeman & Jörg Drechsler. 2026. On the Formal Privacy Guarantees of Synthetic Data (Generated Without Formal Privacy Guarantees), in: *Harvard Data Science Review* Special Issue 6.

14. Arnold, Christian & **Marcel Neunhoeffer**. 2025. Working with Synthetic Data: The Do's, Dangers and Don'ts, in: *Sage Research Methods: Research and Data Literacy*.

13. Nate Breznau, Eike Mark Rinke, Alexander Wuttke [& 182 others, including **Marcel Neunhoeffer**]. 2025. The reliability of replications: a study in computational reproductions. *Royal Society Open Science* 12: 241038.

12. Roni Lehrer, Oke Bahnsen, Klara Müller, **Marcel Neunhoeffer**, Thomas Gschwend & Sebastian Juhl. 2025. Rallying around the leader in times of crises: The opposing effects of perceived threat and anxiety. *European Journal of Political Research* 64 (2): 697-718.

11. Oliver Rittmann, **Marcel Neunhoeffer** & Thomas Gschwend. 2025. How to Improve the Substantive Interpretation of Regression Results when the Dependent Variable is logged. *Political Science Research and Methods* 13 (1): 203-211.

10. Jonathan Latner, **Marcel Neunhoeffer** & Jörg Drechsler. 2024. Generating Synthetic Data Is Complicated: Know Your Data and Know Your Generator. *Lecture Notes in Computer Science, Privacy in Statistical Databases*, International Conference, PSD 2024, Antibes Juan-les-Pins, France, September 25–27, 2024, Proceedings.

9. Mark Bun, Marco Gaboardi, **Marcel Neunhoeffer** & Wanrong Zhang. 2024. Continual Release of Differentially Private Synthetic Data from Longitudinal Data Collections. *Proc. ACM Manag. Data* 2, 2, Article 94 (May 2024), 26 pages.

8. Christian Arnold, Luka Biedebach, Andreas Küpfer & **Marcel Neunhoeffer**. 2024. The Role of Hyperparameters in Machine Learning Models and How to Tune Them. *Political Science Research and Methods* 12 (4): 841-848.

7. Nate Breznau, Eike Mark Rinke, Alexander Wuttke [& 163 others, including **Marcel Neunhoeffer**]. 2022. Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty. *PNAS* 119 (44): 1-8.

6. Thomas Gschwend, Klara Müller, Simon Munzert, **Marcel Neunhoeffer** & Lukas F. Stoetzer. 2022. The Zweitstimme Model: A Dynamic Forecast of the 2021 German Federal Election. *PS: Political Science & Politics* 55 (1): 85-90.

5. **Marcel Neunhoeffer**, Zhiwei Steven Wu & Cynthia Dwork 2021. Private Post-GAN Boosting. *Proceedings 9th International Conference on Learning Representations.*

4. **Marcel Neunhoeffer**, Thomas Gschwend, Simon Munzert & Lukas F. Stoetzer, 2020. An Approach to Predicting the District Vote Shares in German Federal Elections. *Politische Vierteljahresschrift* 61 (1): 111-130.

3. Lukas F. Stoetzer, **Marcel Neunhoeffer**, Thomas Gschwend, Simon Munzert & Sebastian Sternberg. 2019. Forecasting Elections in Multi-Party Systems: A Bayesian Approach Combining Polls and Fundamentals. *Political Analysis* 27 (2): 255-262.

2. **Marcel Neunhoeffer** & Sebastian Sternberg. 2019. How Cross-Validation Can Go Wrong and What to Do About it. *Political Analysis* 27 (1): 101-106.

1. Lukas F. Stoetzer, Simon Munzert, Thomas Gschwend, **Marcel Neunhoeffer** & Sebastian Sternberg. 2017. Ein strukturell-dynamisches Vorhersagemodell für Bundestagswahlen. *Politische Vierteljahresschrift* 58 (3): 418-442.

Other Publications

1. **Marcel Neunhoeffer**. 2018. Book Review: In-Your-Face Politics: The Consequences of Uncivil Media by Diana C Mutz. *Political Studies Review* 16 (1): NP76.

Articles Under Review

Marcel Neunhoeffer, Malte Schierholz, Julia von Bartenwerffer, Jacob Beck, Felix Henninger, Leonhard Kestel, Michael König, Heidi Seibold, Wiebke Weber & Frauke Kreuter. Learning by Solving: Building Government Data Literacy One Problem at a Time.

Marcel Neunhoeffer. Diversity Profiles for Party Systems.

Marcel Neunhoeffer, Jonathan Latner & Jörg Drechsler. When Privacy Measures Fail: A Theoretical Critique of Empirical Disclosure Risk Assessment for Synthetic Data.

Work in Progress

Marcel Neunhoeffer. A Common Benchmark to Evaluate Multiple Imputation Algorithms.

Marcel Neunhoeffer, Daniel Sheldon & Adam Smith. A Bootstrap-based General-purpose Approach for Statistical Inference with Differential Privacy.

Marcel Neunhoeffer, Shlomi Hod & Jörg Drechsler. To Count or Not to Count: Practical DP Mean Estimation with Unknown Dataset Size

Marcel Neunhoeffer, Maria Sirvent & Malte Schierholz. What Information Do Job Embeddings Contain About Occupational Structure? Probing Against the German Occupational Classification

Maria Sirvent & **Marcel Neunhoeffer**. Uncertainty-Aware Embeddings for Occupational Similarity: Evidence from German Labor Market Data

Invited Talks

ITACOSM 25, On the Formal Privacy Guarantees of Synthetic Data (Generated without formal Privacy Guarantees), with Jeremy Seeman & Jörg Drechsler, 2025.

NBER Conference: Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches and their Consequences, On the Formal Privacy Guarantees of Synthetic Data (Generated without formal Privacy Guarantees), with Jörg Drechsler & Jonathan Latner, 2024.

NBER Conference: Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches and their Consequences, A Bootstrap-based General-purpose Approach for Statistical Inference with Differential Privacy, with Daniel Sheldon & Adam Smith, 2023.

Social Science Data Lab, Generative Adversarial Nets for Social Scientists, 2020.

Workshop on Economics of Privacy and Data Labor at ICML, Really Useful Synthetic Data – A Framework to Evaluate the Quality of Differentially Private Synthetic Data with Christian Arnold, 2020.

International Methods Colloquium, Flexible Multiple Imputation for Social Science using Generative Adversarial Nets, 2019.

Conference & Workshop Presentations

Computer Science: The International Conference on Learning Representations (2021), Theory and Practice of Differential Privacy (2020, 2023, 2024), The Symposium on Principles of Database Systems (2024)
Statistics: Joint Statistical Meetings (2022, 2023), DAGStat Conference (2019)
Social Sciences: PolMeth Europe (2021), European Political Science Association General Conference (2017, 2019), Annual PSA Methodology Conference (2019), APSA Annual Meeting and Exhibition (2018), Annual Meeting of the Society for Political Methodology (2018), European Consortium for Political Research General Conference (2017)

Teaching and Supervision

Teaching

Diplomatische Akademie, Vienna
Seminar: Machine Learning
Graduate (Spring 2026)

Ludwig-Maximilians-Universität München, Department of Statistics
Research Seminar: Applied Data Analytics for the Public Sector
Introducing and Improving Data Analysis for Archivists
Graduate and Professional Development (Spring 2024)

Statistical Consulting: Applied Data Analytics for the Public Sector
Implementing a Statistical Sampling Tool for the Bavarian State Archives
Graduate and Professional Development (Fall 2023)

Research Seminar: Data and Privacy?
Theoretical and Empirical Challenges and Strategies
Graduate (Spring 2022)

Lecture: Statistics II for Sociologists
Undergraduate (Spring 2021)

University of Mannheim, School of Social Sciences
Multivariate Analyses
Graduate (Fall 2016, 2017, 2018, 2019, 2020)

Advanced Quantitative Methods
Graduate (Spring 2017, 2018, 2019, 2020)

University of California, Berkeley, School of Information
Privacy Engineering

Graduate (Summer 2020)

University of Applied Sciences Ludwigshafen
Applied Marketing Research
Graduate (Spring 2017)

Professional Teaching Experience
Generative Adversarial Networks
Data Science Summer School 2022
Online/Hertie School Berlin (August 2022)

Introduction to Machine Learning
CIVICA Data Science Summer School
Online/Hertie School Berlin (July 2021)

Big Data and Social Science
GRADE – Goethe Research Academy for Early Career Researchers
Frankfurt am Main (June 2019)

Supervised and unsupervised Machine Learning and Deep Learning
Bundesbank, Frankfurt am Main (March 2019)

Introduction to R
Geschäftsstelle für Qualitätssicherung Hessen, Eschborn
(February 2018)

Supervision

Master Theses
Katharina Julia Brenner, 2024. Didaktische Ansätze für Stichprobentheorie im Archivwesen. Design und Umsetzung einer Schulung in Zusammenarbeit mit der Generaldirektion der Staatlichen Archive Bayerns. *Ludwig-Maximilians-Universität München*.

Outreach, Honors, Service and Skills

Public Outreach

Co-founder and contributor zweitstimme.org, German Federal Election Forecast, 2017–Present.

Thomas Gschwend & **Marcel Neunhoeffer** (23 January 2020):"CSUler in die Container? Oder wie man doch noch zu einem Bundestag mit 598 Abgeordneten kommen kann." [Put the CSU MPs in temporary offices? Or how a Bundestag with 598 members of parliament can be achieved.] Verfassungsblog.

Thomas Gschwend, **Marcel Neunhoeffer** & Marie-Lou Sohnius (09 August 2019): "Grünes Umfragehoch - Warum der Hype diesmal bis zur nächsten Wahl dauern könnte" [Green Polling High - Why the hype could carry over to the next election this time.] Tagesspiegel.

Thomas Gschwend, Thomas König & **Marcel Neunhoeffer** (14 October 2018): "Wie das bayerische Wahlrecht die CSU begünstigt." [How the electoral rules in Bavaria favor the CSU.] Zeit Online.

Thomas Gschwend, Simon Munzert, **Marcel Neunhoeffer**, Sebastian Sternberg & Lukas F. Stoetzer (23 September 2017): "New German election forecast: Merkel's party will win but lose seats." The Washington Post, Monkey Cage.

Sebastian Sternberg & **Marcel Neunhoeffer** (01 September 2017): "Das Rennen um Platz drei bleibt spannend." [The race for the third place is a tight one.] Tagesspiegel, Causa.

Honors and Awards

IPID4all Travel Grant, University of Mannheim, Funding for the Research Stay at University of California, Berkeley (3,739 EUR), 2018.

Publication of the Year Award by the CDSS and GESS at the University of Mannheim for "How Cross-Validation Can Go Wrong and What to Do About it" (1,000 EUR) with Sebastian Sternberg, 2018.

Professional Service

Referee Service: *Journal of Clinical and Translational Science, Public Opinion Quarterly, Harvard Data Science Review, American Political Science Review, British Journal of Political Science, Journal of Politics, Party Politics, Political Analysis, Political Science Research and Methods, Politische Vierteljahresschrift, International Journal of Forecasting, Theory and Practice of Differential Privacy*

Member of the Organising Committee for PolMeth Europe 2021

Skills

Software: R, Python, PyTorch, Tensorflow, Keras, Stan; html

Code Repository: github.com/mneunhoe

Languages

German (native), English (fluent), French (basic)